

demonstrated by Taiwan Semiconductor Manufacturing Company.

Multiple fin-height FinFETs. The most advanced semiconductor technology used today can produce around 10 million transistor elements in a chip of size one square centimetre. A new multiple fin-height FinFET process technology has been developed at National Nano Device Laboratories (NDL), National Applied Research Laboratories (NARLabs). It allows up to 20 million transistors in the same place, which is twice the number possible today!

Unlike the traditional field-effect transistors (FETs), where gate controls source-drain current only on one side, the fin field-effect transistor (FinFET) has three-dimensional fork-shaped gate similar to a fish's fin (hence the name). It can control the source-drain current in the transistor from three sides—left, right and top. In multiple fin-height FinFET, like the name suggests, the fins will have different heights allowing different amount of currents corresponding to associated channel height. This concept lets further reduction in chip area by 20%. Compared to a normal FinFET, the multiple-height FinFET has 20% higher storage capacity and 20% less manufacturing cost.

FD-SOI. STMicroelectronics, along with its partners, introduced a technology known as Fully Depleted Silicon On Insulator (FD-SOI). It is a planar process technology where an ultra-thin layer of insulator (buried oxide or BOX) is positioned on top of the base silicon and then a very thin silicon film implements the transistor channel.

IC etched on graphene. Several manufacturing units have been working on moving from silicon to other alternatives, to get better performance

MAJOR CONTRIBUTORS TO THIS STORY



Bhaskar Dhandapani
country sales manager,
Freescale Semiconductor
India Pvt Ltd



Rajiv Mantri
analog applications
manager, Texas
Instruments (India)



Sanjeev Sharma
director (IP - sales &
marketing), Terminus
Circuits



Somsubhra Pal Choudhury
managing director,
Analog Devices India



Vishal Goyal
senior technical
marketing manager,
STMicroelectronics

in terms of power, size and cost. These alternates could be based on graphene, nanotubes or various tunnel field-effect transistors (TFETs).

There is on-going research on these and other devices based on electron spin as opposed to electron charge. Researchers at University of California, Santa Barbara, have come up with the concept and demonstration of an integrated circuit design scheme where the transistors and interconnects are monolithically patterned seamlessly (without obvious joints or seams) on a two-dimensional atomically flat sheet of graphene.

The model of etching offers possibilities for extremely energy-efficient, transparent and flexible next-generation green electronics.

Photonic IC that can manipulate atoms. A group of scientists from California State University, Pasadena, claim to have developed an optical circuit that uses light to manipulate individual atoms in small, self-contained units. According to them, the photonic integrated circuit is an integrated optical circuit with a photonic crystal that can both localise and interface atoms with guided photons in the device.

The optical properties of photonic crystals are determined by factors like their physical geometry and size of the

waveguide. This feature allowed the scientists to make a photonic crystal of silicon nitride that acts as waveguide for laser light tuned to certain atomic transitions in caesium. The caesium atom absorbs and scatters these wavelengths, generating forces that could trap and manipulate the atom.

The photonic crystal is integrated into a small, self-contained system that has a readily available supply of caesium atoms. These units are relatively cheap to make and easy to operate. This interesting technology could serve as high-quality building block for quantum computation and communication. It also opens door for several experimental opportunities.

Analogue standing tall

Over the last few years, we have witnessed the shifting of electronics world from analogue to more sophisticated and efficient digital substitutes. But all analogue components cannot be completely substituted by digital devices; they would coexist and complement each other. With the extensive developments in the IoT segment, we see it driving many new technologies and products in analogue components and chips. ●

The author is a technical correspondent at EYF



For More
Test and Equipment Stories
Visit www.electronicstest.com

